colors and now this technique is equally well applied to separation of colorless substances.

The principle behind chromatography is selective distribution of the mixture of organic substances between two phases – a stationary phase and a moving phase. The stationary phase can be a solid or liquid, while the moving phase is a liquid or a gas. When the stationary phase is a solid, the moving phase is a liquid or a gas. If the stationary phase is solid, the basis is adsorption, and when it is a liquid, the basis is partition. So the Chromatography is defined as a technique for the separation of a mixture brought about by differential movement of the individual compound through porous medium under the influence of moving solvent. The various methods of chromatography are

1. Column chromatography (CC)
2. Thin layer chromatography (TLC)
3. Paper chromatography (PC)
4. Gas-liquid chromatography (GLC)
5. Ion–exchange chromatography

**Adsorption chromatography:** The principle involved is different compounds are adsorbed on an adsorbent to different degree. Silica gel and alumina are the commonly used adsorbent. The components of the mixture move by varying distances over the stationary phase. Column chromatography and thin layer chromatography are the techniques based on the principle of differential adsorption.

**Column chromatography:** This is the simplest chromatographic method carried out in long glass column having a stop cock near the lower end. This method involves separation of a mixture over a column of adsorbent (Stationery phase) packed in a column. In the column a plug of cotton or glass wool is placed at the lower end of the column to support the adsorbent powder. The tube is uniformLy packed with suitable absorbent constitute the stationary phase. (Activated aluminum oxides (alumina), Magnesium oxide, starch are also used as absorbents).

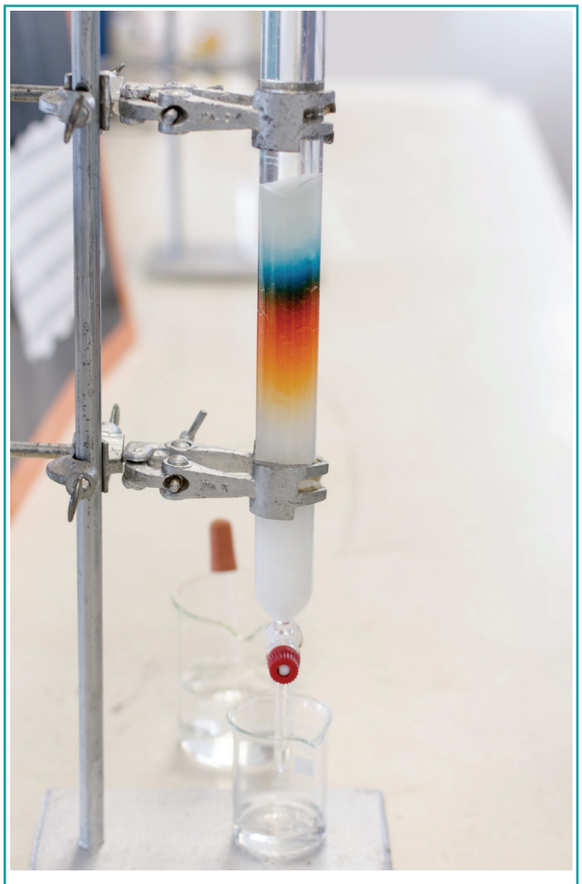


Fig 11.5 Column Chromatography

The mixture to be separated is placed on the top of the adsorbent column. Eluent which is a liquid or a mixture of liquids is allowed to flow down the column slowly. Different components are eluted depending upon the degree to which the components are adsorbed and complete separation takes place. The most readily adsorbed substances



11th Chemistry Chapter-11 Vol-2 EM.indd 152 09-01-2023 14:43:16